

Internship Project Report



AIRPORTS AUTHORITY OF INDIA

Submitted by,

Shantil Khadatkar

Mukesh Patel school of technology management & engineering,
Mumbai

Under The Guidance of,

S V Manohj

Manager (ATM)

Saurabh Khatri

Assistant Manager (ATM)

Index

Sr. No.	Name
1.	Introduction
2.	Flight Progress Strip data Extraction using PaddleOCR model - python
3.	Strips Information Extraction
4.	Conclusion
5.	Token of Thanks

Introduction

The Airport Authority of India (AAI), established in 1995, is a premier public-sector organization responsible for overseeing and managing civil aviation infrastructure in India. Functioning under the Ministry of Civil Aviation, AAI plays an instrumental role in the development and management of the country's airports, air traffic control services, and air navigation systems. The body is responsible for creating, upgrading, and maintaining world-class aviation facilities that facilitate safe, efficient, and environmentally sustainable operations.

AAI is a leader in modernizing the aviation sector, ensuring that India remains connected to both domestic and international destinations. As one of the largest aviation authorities globally, AAI operates over 137 airports in India, which includes international hubs such as Indira Gandhi International Airport (Delhi), Chhatrapati Shivaji Maharaj International Airport (Mumbai), and Kempe Gowda International Airport (Bangalore). These airports serve millions of passengers annually, contributing significantly to the economic growth of India by promoting tourism, trade, and business development.

Vision and Mission of AAI

Vision:

AAI envisions itself as a world-class organization that is committed to providing safe, secure, and efficient air traffic management while ensuring that the airport infrastructure supports sustainable development. The vision extends to promoting regional connectivity and improving air navigation services by employing state-of-the-art technologies and best practices.

Mission:

AAI's mission is rooted in its endeavour to modernize airports, provide seamless connectivity, and deliver world-class aviation services. A key component of AAI's mission is the UDAN Scheme (Ude Desh ka Aam Naagrik), which aims to make air travel accessible and affordable for every citizen. Under this mission, AAI also seeks to enhance customer experience by delivering world-class amenities, safety protocols, and environmental sustainability at all airports.

Key Functions and Responsibilities of AAI

1. Airport Development and Maintenance:

AAI is tasked with developing, upgrading, and maintaining airports across the country. The authority's efforts focus on expanding the capacity of airports to accommodate increasing passenger traffic, along with enhancing infrastructure to meet global standards. This includes runway construction, terminal expansions, and modernization of passenger facilities.

2. Air Traffic Management (ATM):

AAI manages air traffic control services across the Indian airspace and adjoining oceanic areas. This includes providing safe and efficient air navigation services, coordinating flight movements, and ensuring that air traffic is managed with the highest levels of precision and security. The ATM services aim to minimize delays, optimize flight paths, and reduce carbon emissions.

3. Passenger Services and Experience:

AAI focuses on improving the passenger experience by offering modern, comfortable, and efficient airport services. This includes faster check-in procedures, improved security screening, and providing world-class lounges, duty-free shopping, and hospitality services. The objective is to enhance the passenger journey from arrival to departure while prioritizing safety and convenience.

4. Sustainability Initiatives:

AAI is committed to adopting eco-friendly practices and integrating sustainable technologies into airport operations. Green airport initiatives focus on reducing energy consumption, managing waste responsibly, and minimizing environmental impact through the implementation of solar energy, rainwater harvesting, and carbon-neutral initiatives.

5. Promoting Regional Connectivity:

Through the UDAN scheme, AAI ensures that underserved regions are connected to major airports, promoting inclusive economic growth. By developing airports in Tier-2 and Tier-3 cities, AAI fosters accessibility and creates new opportunities for local communities.

Departments of AAI

1. Data Management Systems (DMS) Department

The Data Management Systems (DMS) department at AAI plays a pivotal role in managing, analyzing, and utilizing the vast amounts of data generated across various airport operations. This department focuses on collecting, organizing, and processing data related to flights, passenger movements, and airport infrastructure. DMS utilizes advanced analytics and reporting tools to generate insights that facilitate effective decision-making. One of the key responsibilities of the DMS department is to enhance operational efficiency by ensuring data accuracy and integrity. By leveraging data analytics, DMS can identify trends in flight operations, monitor performance metrics such as on-time departures and arrivals, and detect anomalies that may require corrective action. Furthermore, the DMS department collaborates with other departments to create integrated data solutions, ensuring that all stakeholders have access to real-time information for effective operational management.

2. Airport Collaborative Decision Making (ACDM) Department

The Airport Collaborative Decision Making (ACDM) department is crucial in enhancing the efficiency and effectiveness of airport operations by fostering collaboration among various stakeholders. ACDM focuses on real-time information sharing and decision-making processes between airlines, ground handling services, air traffic controllers, and airport operators. By implementing ACDM principles, AAI aims to minimize delays, optimize resource allocation, and improve passenger experience. The department uses advanced technologies and data analytics to create a shared situational awareness among all parties involved in flight operations. ACDM initiatives lead to more informed decisions regarding runway usage, gate assignments, and ground handling, ultimately enhancing the overall throughput of airport operations. Additionally, ACDM engages in continuous improvement efforts by analyzing performance data to refine processes and enhance coordination among stakeholders.

3. Air Traffic Control (ATC) Department

The Air Traffic Control (ATC) department is responsible for ensuring the safe and efficient movement of aircraft both in the airspace and on the ground. This department plays a critical role in managing air traffic by providing instructions to pilots regarding takeoff, landing, and in-flight navigation. The ATC operates from various control towers located at airports and area control centers that oversee large sections of airspace. Air traffic controllers utilize advanced radar systems, communication technologies, and air traffic management tools to monitor flight paths and manage aircraft separation. The department also plays a key role in emergency situations, providing guidance and support to pilots in distress. ATC collaborates closely with other departments to ensure that air traffic is managed smoothly, minimizing delays and maintaining safety standards.

4. Engineering Department

The Engineering Department at AAI is tasked with the planning, design, construction, and maintenance of airport infrastructure. This department ensures that all facilities, including runways, taxiways, terminals, and navigational aids, meet international safety and operational standards. The engineering team is involved in the implementation of modernization projects, including the installation of advanced technologies and systems to enhance airport operations. They also conduct regular inspections and maintenance to ensure the integrity and safety of airport infrastructure. The Engineering Department collaborates with other departments to address any infrastructure-related challenges and implements innovative solutions to improve the overall efficiency and safety of airport operations.

5. Commercial and Business Development Department

The Commercial and Business Development Department focuses on enhancing the financial performance of AAI by optimizing revenue generation through commercial activities. This department is responsible for managing airport concessions, retail spaces, and advertising contracts. The team identifies opportunities for business partnerships and collaborations that can enhance the overall passenger experience while maximizing non-aeronautical revenue streams. The Commercial Department also plays a critical role in strategic planning, conducting market research to understand passenger preferences and trends. By fostering relationships with airlines, businesses, and service providers, the department aims to create a vibrant and profitable airport environment.

6. Human Resources Department

The Human Resources (HR) Department at AAI is responsible for managing the organization's workforce, focusing on recruitment, training, employee development, and performance management. HR plays a vital role in ensuring that AAI has a skilled and motivated workforce to meet the challenges of the rapidly evolving aviation sector. The department develops policies and programs that promote employee engagement, diversity, and inclusion. HR also handles employee relations, ensuring a positive work environment while addressing grievances and concerns. Additionally, the department conducts training and development programs to enhance the skills and competencies of employees, aligning them with the organization's goals and objectives.

7. Finance Department

The Finance Department is integral to the financial health of AAI, managing budgeting, financial planning, accounting, and reporting activities. This department ensures that resources are allocated efficiently and that financial practices comply with regulatory requirements. The Finance team monitors the organization's financial performance, conducting analyses that inform strategic decision-making. They are responsible for managing revenue streams, including aeronautical and non-aeronautical income, and preparing financial reports that provide insights into operational efficiency and profitability. Additionally, the Finance Department plays a critical role in forecasting future financial trends, which helps AAI in strategic planning and investment decisions.

8. Security Department

The Security Department at AAI is tasked with ensuring the safety and security of airport operations, passengers, and staff. This department implements and enforces security protocols and procedures in compliance with national and international regulations. The Security team conducts regular risk assessments and audits to identify vulnerabilities and mitigate potential threats. They are responsible for screening passengers and baggage, monitoring airport premises, and coordinating emergency response measures in case of security incidents. The department collaborates closely with law enforcement agencies and other stakeholders to maintain a secure environment for all airport activities.

Importance of AAI in India's Growth

AAI plays a critical role in the nation's economy by supporting the growth of aviation, tourism, and trade. The development of international and regional airports has helped bridge geographical barriers, making India more accessible. AAI's commitment to modernizing airport infrastructure and air traffic management is key to positioning India as a global aviation leader. Moreover, by implementing sustainable practices, AAI contributes to environmentally friendly aviation operations, which are vital in the context of climate change.

ACDM Flight Progress Strip data Extraction using PaddleOCR model python

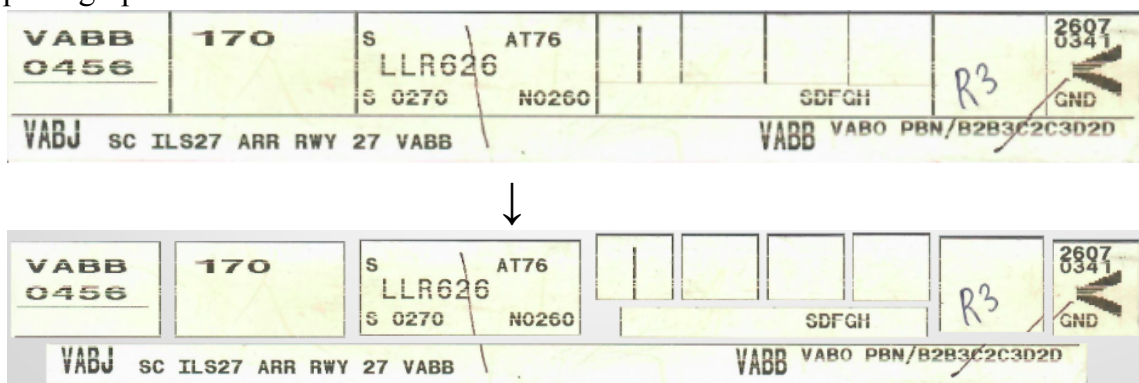
During my internship at AAI, I had the opportunity to work closely with the ACDM department. The primary focus of my assignment was to extract flight progress data, specifically so that we can extract its contents and digitalize the process. This section of my internship offered me a deep dive into the workings of paddleocr module and using flask to passthrough image and host webpage in the network.

Understanding the Technical challenge

The primary technical challenge of this project was to automate the extraction of data from physical flight progress strips. These strips, while essential for air traffic control, are physical documents that require manual data entry to be digitized. This process is not only time-consuming but also prone to human error. The goal was to create a system that could take an image of a flight progress strip, automatically identify and extract the relevant information, and present it in a structured digital format.

To address this challenge, I developed a solution that leverages several key technologies:

- **Image Preprocessing with OpenCV:** Before any text can be extracted, the image of the flight progress strip must be processed to ensure its quality and to isolate the relevant data fields. I used the OpenCV library in Python to perform a series of image transformations. This included converting the image to grayscale, applying Gaussian blur to reduce noise, and using Canny edge detection to identify the borders of the strip. I then used contour detection and a perspective transform to "flatten" the image, correcting for any skew or distortion from the angle of the photograph.



- **Optical Character Recognition (OCR) with PaddleOCR:** Once the image was preprocessed, I used the PaddleOCR engine to perform the actual text extraction. PaddleOCR is a powerful, open-source OCR toolkit that is well-suited for this task due to its high accuracy and support for various languages and text orientations. The preprocessed image was divided into smaller segments, each corresponding to a specific data field on the flight progress strip (e.g., aircraft type, route, etc.). PaddleOCR was then run on each of these segments to extract the text.

- **Web Interface with Flask:** To make the system accessible and easy to use, I developed a simple web interface using the Flask framework in Python. This interface allows a user to upload an image of a flight progress strip directly from their web browser. The backend Flask application then processes the image using OpenCV and PaddleOCR pipeline and displays the extracted data on the webpage.

Upload an Image for OCR

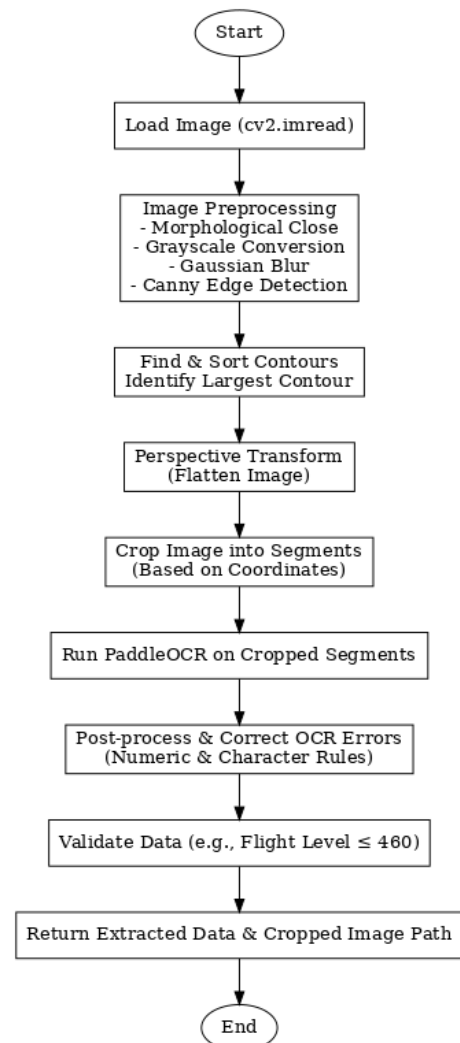
Choose file | No file chosen | Upload

OCR Results:

Departure Aerodrome: VA88
Estimated off-block time (UTC): 0500
Flight levels: 350
Category of Flight: S
Type of Aircraft: A321
Call Sign: A1C809
Squak Mode: 5
Squak Reading: 0314
Speed: N0463
Standard Instrument Departure (SID): ANOL12AX15R15 SDF6H
Strip - Date: 2509
Strip - Time: 0417
Strip - Printed At:
Arriving From: VA88
Standard Instrument Departure (SID): ANOL12AX15R15
Destination Aerodrome: V1DP
Alternate Aerodrome: V1AF
RNAV/RNP specifications: PBN/A1B1C1D1025

Cropped Image:

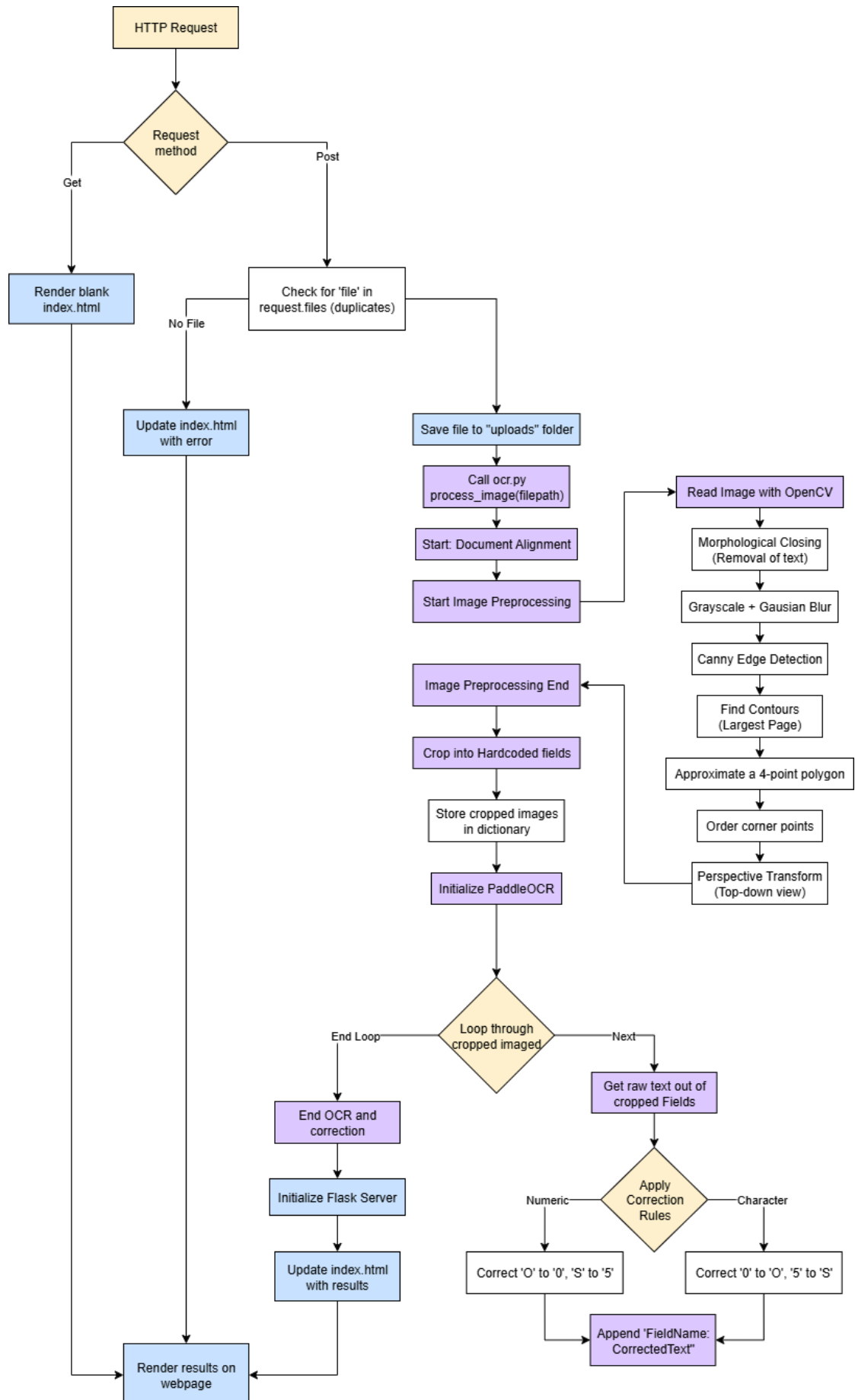
Original Image:



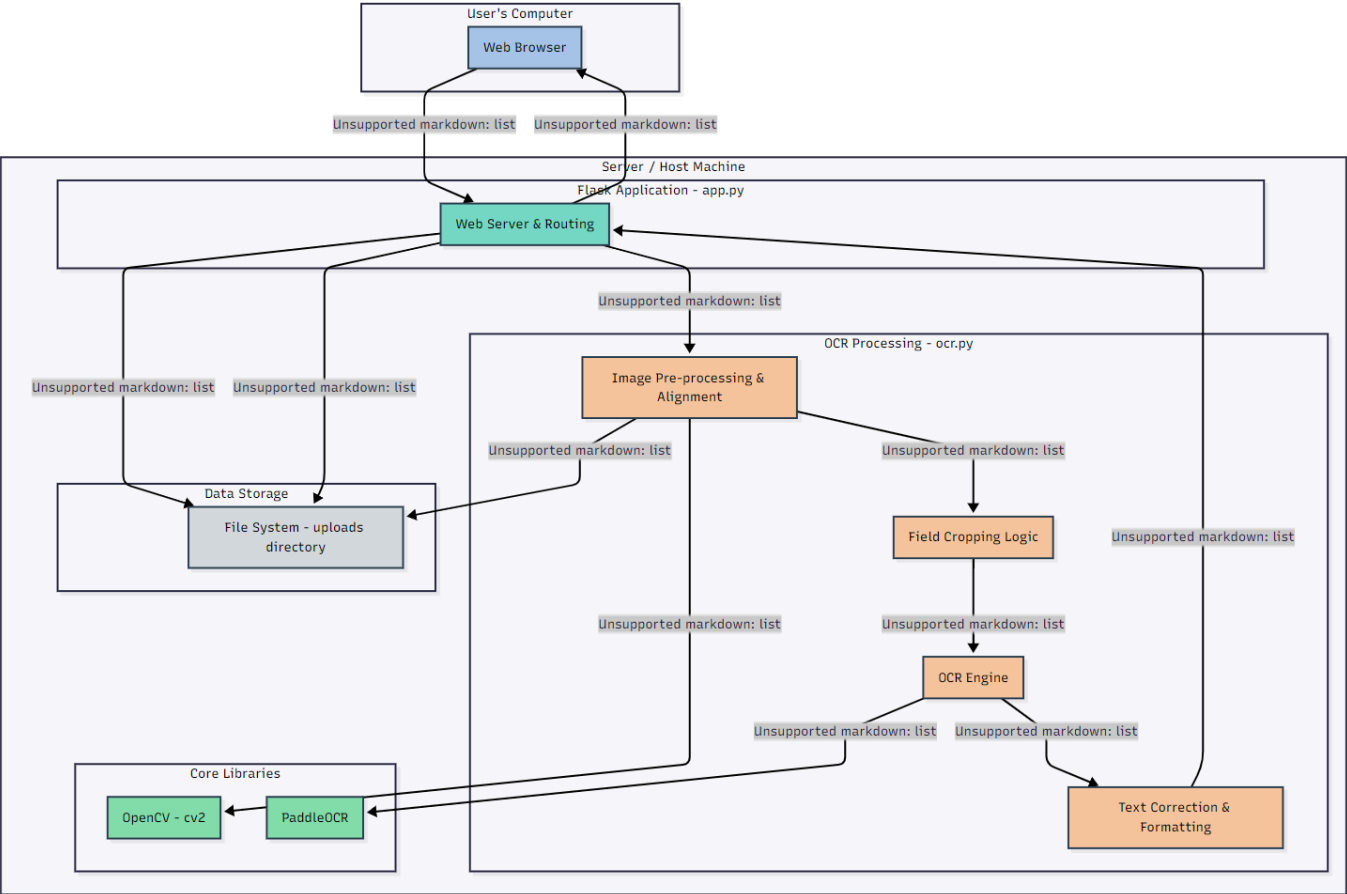
- **Data Correction and Validation:** OCR is not always perfect, so I implemented a post-processing step to correct common OCR errors. For example, the character 'O' is often mistaken for the number '0', and 'I' can be mistaken for '1'. I created a set of rules to correct these and other common errors based on the expected format of the data in each field (e.g., some fields should be numeric, while others should be alphabetic).

By combining these technologies, I was able to create a robust and effective solution for digitizing flight progress strips, significantly reducing the need for manual data entry and improving the accuracy and efficiency of the data collection process.

Below is the Decision Making diagram



Below is Block Diagram representing the same



My Role and Contributions

During my internship, I was responsible for the end-to-end development of the flight progress strip data extraction tool. My contributions can be broken down into the following key areas:

1. System Design and Technology Selection:

I began by researching and evaluating different technologies for each component of the system. I chose OpenCV for its extensive image processing capabilities, PaddleOCR for its high-accuracy OCR engine, and Flask for its lightweight and flexible web framework. I designed the overall architecture of the system, ensuring that each component was modular and could be easily tested and maintained.

2. Image Processing and OCR Pipeline Development:

I wrote the core Python scripts for the image processing and OCR pipeline. This involved:

- Implementing the image preprocessing steps using OpenCV, including grayscale conversion, blurring, edge detection, and perspective correction.
- Developing the logic for cropping the preprocessed image into smaller segments corresponding to the different data fields on the flight progress strip.
- Integrating the PaddleOCR engine to extract text from each of these segments.
- Creating a set of post-processing rules to correct common OCR errors and validate the extracted data.

3. Web Application Development:

I developed the Flask web application that serves as the user interface for the system. This included:

- Writing the backend Python code to handle file uploads, process the images, and render the results.
- Creating the frontend HTML template to display the file upload form, the extracted data, and the processed images.

4. Testing and Debugging:

I was responsible for testing the system with a variety of different images to ensure its accuracy and robustness. I used a set of sample images of flight progress strips to test the image processing and OCR pipeline, and I manually verified the accuracy of the extracted data. I also debugged and fixed any issues that arose during the development and testing process.

Through this project, I gained valuable hands-on experience in computer vision, optical character recognition, and web development. I was able to successfully deliver a functional prototype that demonstrated the feasibility of automating the data extraction process for flight progress strips, and I am confident that this work will provide a solid foundation for future development in this area.

Key Responsibilities

- **Image Preprocessing and Document Scanning**

The first critical step was to process the raw images of the flight progress strips. Since these images could be taken from various angles and under different lighting conditions, I used the OpenCV library to programmatically clean and standardize them. This involved:

- Applying morphological operations to remove noise and isolate the paper strip from the background.
- Converting the image to grayscale and using Canny edge detection to identify the boundaries of the strip.
- Implementing a contour detection algorithm to find the four corners of the document within the image.
- Applying a perspective transform to correct for any skew, effectively creating a "flattened," top-down view of the strip as if it had been perfectly scanned. This step was crucial for ensuring the accuracy of the subsequent OCR process.

- **Data Cleaning and Validation**

Recognizing that OCR technology is not always perfect, a significant part of my role was to clean and validate the extracted data to ensure its integrity. I developed a custom function to post-process the OCR output, which included:

Implementing rules to correct common character misinterpretations, such as mistaking the letter 'O' for the number '0', 'I' for '1', or 'S' for '5', based on whether the field was expected to be numeric or alphabetic.

Enforcing data validation rules based on operational knowledge. For example, I added a check to flag when a flight level value exceeded a realistic maximum (e.g., 460), alerting the user to a potential OCR error.

- **Reporting and Visualization**

To make the system user-friendly and the results easily accessible, I was responsible for creating a web-based interface using Flask. This involved:

- Building a simple web page where users could upload an image of a flight progress strip.
- Developing the backend logic to orchestrate the entire image processing and OCR pipeline.
- Presenting the results clearly to the user, displaying the original uploaded image, the perspective-corrected image, and a structured list of the extracted and cleaned data fields. This provided immediate visual feedback and allowed for easy verification of the results.

Outcome of My Work

The successful development of this OCR tool had a significant and tangible impact on the ACDM department's workflow. By creating an automated system to digitize flight progress strips, my work directly addressed the inefficiencies and potential for error inherent in the manual data entry process.

The primary outcome was a functional prototype that could take a simple photograph and, within seconds, produce a structured, digital record of the flight data. This provided critical insights that enabled the team to:

- **Enhance Operational Efficiency:** The tool drastically reduced the time and manual effort required to digitize flight data, freeing up personnel to focus on more analytical tasks.
- **Improve Data Accuracy:** By automating the extraction and applying programmatic correction rules, the system minimized the human errors that can occur during manual transcription, leading to more reliable and consistent data.
- **Create a Foundation for Advanced Analytics:** The extracted digital data can now be easily stored and fed into larger data analysis systems. This paves the way for future projects, such as analyzing flight patterns, tracking on-time performance, and identifying operational bottlenecks on a much larger scale, directly contributing to the data-driven optimization goals of the AAI.

Ultimately, this project served as a successful proof-of-concept for how computer vision and machine learning can be applied to solve real-world challenges in air traffic management. It not only provided an immediate benefit to the ACDM department but also contributed to the broader goal of creating a more efficient, accurate, and sustainable aviation ecosystem in India.

Conclusion

My internship at the Airport Authority of India (AAI) provided an invaluable opportunity to apply technical skills to a real-world aviation challenge while gaining insight into the complex operations of one of the nation's most critical public sector organizations. Working with the Airport Collaborative Decision Making (ACDM) department, I was able to design and implement an end-to-end solution for automating the extraction of data from flight progress strips using OpenCV for image preprocessing, PaddleOCR for text recognition, and Flask for building a user-friendly web interface.

The project successfully demonstrated that manual, error-prone processes in air traffic management can be streamlined through the integration of computer vision, OCR, and web-based automation tools. The developed prototype significantly reduced the time required for data entry, improved accuracy through programmatic error correction, and laid the groundwork for advanced data analytics applications in the future.

This experience not only enhanced my knowledge of image processing, OCR pipelines, and web development but also deepened my understanding of operational efficiency within aviation. The collaborative work with AAI professionals allowed me to bridge the gap between technical implementation and industry-specific requirements.

Ultimately, the internship reinforced the value of leveraging emerging technologies to address operational challenges, improve accuracy, and enable data-driven decision-making. The skills, professional exposure, and practical problem-solving experience I gained will continue to guide my future academic and career endeavors in technology-driven solutions for critical infrastructure systems.

Token of Thanks

I would like to express my deepest gratitude to the Airport Authority of India (AAI) for providing me with this exceptional internship opportunity. My experience during this internship has been profoundly enriching, both professionally and personally, and I owe this to the supportive and knowledgeable individuals I had the privilege of working with. I am particularly thankful to ACDM departments for their guidance and collaboration, which played a pivotal role in helping me understand the operational intricacies of air traffic management. Their patience and expertise in mentoring me through the various projects allowed me to contribute meaningfully while expanding my own skill set.

I would also like to extend a heartfelt thanks to my mentors and supervisors at AAI, whose valuable insights and continuous support were instrumental in guiding me throughout my internship. They provided me with the freedom to explore innovative solutions, while also offering constructive feedback that helped me refine my technical abilities. Their commitment to nurturing young talent like mine, coupled with their dedication to upholding the highest standards of professionalism, truly made this experience an unforgettable learning journey.

I am immensely grateful for the opportunity to work on impactful projects that directly contribute to improving the efficiency and sustainability of India's aviation sector. The hands-on experience I gained from using data analytics, machine learning, and image